MOBILE COMPUTING LAB / NETWORK OBSERVATORY

MOBILE NETWORK PROTOCOL STACK

• OSI REFERENCE / STATIC STUDY VIEW

Understanding the layers that enable wireless communication

01 / END-TO-END COMMUNICATION

■ DATA PACKET / BIDIRECTIONAL TRANSFER



02 / LAYERED ARCHITECTURE

7 LAYERS / OSI REFERENCE MODEL

7	Application	Web, email, video, and mobile apps.
6	Presentation	Data formatting, encryption, and compression.
5	Session	Session establishment, management, and synchronization.
4	Transport	Reliable and best-effort delivery using TCP and UDP.
3	Network	IP addressing and routing using IPv4/IPv6.
2	Data Link	Framing, MAC addressing, and error detection.
1	Physical	Transmission of bits through radio waves, LTE, 5G, and Wi-Fi.

ⓘ OSI reference model applied to mobile networking; not a literal seven-layer LTE/5G protocol stack.

01

LAYER 1 / OSI REFERENCE MODEL

Physical

BITS, SYMBOLS & THE RADIO CHANNEL

THE IDEA, IN PLAIN LANGUAGE

Transmission of bits through radio waves, LTE, 5G, and Wi-Fi. At the physical layer, a radio turns coded bits into signals and recovers them from the received waveform.

STACK CONTEXT

7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data Link
1 Physical •

DATA UNIT
Bits / radio symbols

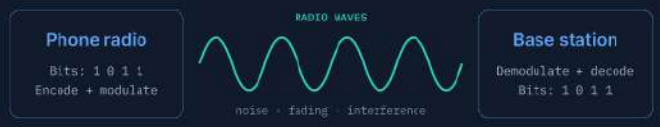
SEND ↓ / RECEIVE ↑

01 / HOW IT WORKS

CONCEPTUAL DATA FLOW

From bits to a signal in the air

Radio modulation maps coded bits onto symbols. The receiver measures the signal and estimates the transmitted bits.



BITS → CODED SYMBOLS → RF SIGNAL → ESTIMATED BITS

ILLUSTRATIVE WAVEFORM / NOT AN ACTUAL LTE OR NR SIGNAL

FOLLOW THE DATA

Send: Layer 2 provides bits; the radio codes and modulates them for the channel. Receive: demodulation and decoding recover bits that are passed back to Layer 2.

02 / CORE RESPONSIBILITIES

01 **Encode and modulate**
Map coded bits to radio symbols and generate a waveform.

02 **Use physical resources**
Operate on assigned frequencies, timing, bandwidth, and power.

03 **Recover received bits**
Estimate symbols and decode despite noise and fading.

EXAMPLE PROTOCOLS / STANDARDS

LTE PHY	4G radio physical-layer procedures.
5G NR PHY	5G radio physical-layer procedures.
802.11 PHY	Wi-Fi physical-layer specifications.

LTE, 5G NR, and Wi-Fi span multiple layers. Only their PHY functions belong here; the radio carries encoded symbols, not app messages directly.

03 / ON YOUR PHONE

Sending a photo on 5G

After a photo's data has passed through higher layers, the phone's 5G NR radio sends coded symbols over assigned radio resources. The gNodeB estimates those symbols and decodes the bits for further processing.

REMEMBER

Layer 1 asks: how do the bits cross the physical medium?

OSI responsibilities applied to mobile networking — not a literal seven-layer LTE/5G implementation.

02

LAYER 2 / OSI REFERENCE MODEL

Data Link

LOCAL-LINK DELIVERY

THE IDEA, IN PLAIN LANGUAGE

Framing, MAC addressing, and error detection. This layer organizes network packets into frames for transfer over one local link, such as a phone's Wi-Fi connection.

STACK CONTEXT

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network
- 2 Data Link •
- 1 Physical

DATA UNIT
Frame

SEND ↓ / RECEIVE ↑

01 / HOW IT WORKS

CONCEPTUAL DATA FLOW

Wrap a packet for the next local hop

A simplified Wi-Fi frame carries local addressing, an IP payload, and an error-detection field. Its scope is the link, not the whole internet.



FOLLOW THE DATA

Send: encapsulate an IP packet into a link frame, then hand bits to the physical layer. Receive: detect frame errors, remove link fields, and deliver the packet up.

02 / CORE RESPONSIBILITIES

01 **Frame and address locally**
Mark frame boundaries and identify local-link interfaces.

02 **Coordinate medium access**
Control how devices share a wireless channel or other medium.

03 **Detect transmission errors**
Use a frame check sequence; some links also retry failed frames.

EXAMPLE PROTOCOLS / STANDARDS

802.11 MAC	Wi-Fi framing and channel access.
802.3 MAC	Ethernet framing and addresses.
LTE / NR L2	MAC, RLC, PDCP; NR also uses SDAP.

MAC is a sublayer, not a single protocol. Wi-Fi may use multiple address fields; cellular Layer 2 does not simply reuse Wi-Fi MAC addressing.

03 / ON YOUR PHONE

Sending over campus Wi-Fi

A phone places an IP packet inside an 802.11 frame for the campus access point. The receiver checks the FCS for corruption. The router later forwards the IP packet inside a new frame on its next link.

REMEMBER

Layer 2 delivers across one link; Layer 3 carries packets across networks.

OSI responsibilities applied to mobile networking — not a literal seven-layer LTE/5G implementation.

03

LAYER 3 / OSI REFERENCE MODEL

Network

LOGICAL ADDRESSING & ROUTING

THE IDEA, IN PLAIN LANGUAGE

IP addressing and routing using IPv4/IPv6. This layer moves packets across interconnected networks, choosing the next hop toward the destination IP address.

STACK CONTEXT

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network •
- 2 Data Link
- 1 Physical

DATA UNIT
IP packet

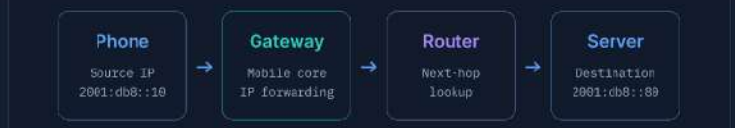
SEND ↓ / RECEIVE ↑

01 / HOW IT WORKS

CONCEPTUAL DATA FLOW

One destination, several network hops

Each router forwards the packet using its routing table. A mobile access network connects the phone to the wider IP path.



IP PACKET DST 2001:db8::80 | transport payload

SIMPLIFIED PATH / DOCUMENTATION ADDRESSES

FOLLOW THE DATA

Send: IP wraps transport data in a packet with source and destination addresses. Routers select next hops; the destination host passes the payload up to transport.

02 / CORE RESPONSIBILITIES

01 **Assign logical addresses**
IPv4 or IPv6 identifies the source and destination interfaces.

02 **Route across networks**
Use destination prefixes and routing tables to choose next hops.

01 **Limit and diagnose forwarding**
TTL / hop limit bounds the path; ICMP reports network conditions.

EXAMPLE PROTOCOLS / STANDARDS

IPv4	32-bit logical addressing.
IPv6	128-bit logical addressing.
ICMP / v6	Error reports and diagnostics.

IP is best-effort: it does not guarantee delivery or ordering. Cellular tunnels, address translation, and mobility make real paths more complex than this diagram.

03 / ON YOUR PHONE

Reaching a college server

A phone sends an IP packet through the mobile access network and core toward a college website. Routers inspect the destination IP prefix and forward it hop by hop; the web server receives the packet at the far end.

REMEMBER

Layer 3 asks: which network destination, and which next hop?

OSI responsibilities applied to mobile networking — not a literal seven-layer LTE/5G implementation.

04

LAYER 4 / OSI REFERENCE MODEL

Transport

PROCESS-TO-PROCESS DELIVERY

THE IDEA, IN PLAIN LANGUAGE

Reliable and best-effort delivery using TCP and UDP. This layer moves data between application endpoints and uses port numbers to deliver it to the right process.

STACK CONTEXT

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport •
- 3 Network
- 2 Data Link
- 1 Physical

DATA UNIT

TCP segment / UDP datagram

SEND ↓ / RECEIVE ↑

01 / HOW IT WORKS

CONCEPTUAL DATA FLOW

Two delivery styles, different trade-offs

TCP recovers losses and delivers an ordered byte stream. UDP sends independent datagrams without those guarantees.

TCP 01 02 03 → ACK + retransmit

Ordered, reliable byte stream

UDP 01 × 03 → No built-in retry

Loss may be handled by the app

PHONE : 53124 → SERVER : 443 / EXAMPLE TCP ENDPOINTS

FOLLOW THE DATA

Send: transport adds ports and TCP or UDP control information before passing data to IP. Receive: it delivers the payload to the matching application endpoint.

02 / CORE RESPONSIBILITIES

01 **Identify application endpoints**
Use source and destination ports to multiplex app traffic.

02 **Choose delivery guarantees**
TCP orders bytes and retransmits losses; UDP does not.

03 **Control the sending rate**
TCP uses flow and congestion control to regulate transmission.

EXAMPLE PROTOCOLS / STANDARDS

TCP	Connection-oriented reliable stream.
UDP	Best-effort datagram delivery.
QUIC	Reliable secure streams over UDP.

UDP is not inherently faster in every case. QUIC adds reliability and congestion control above UDP, showing that internet protocols can span responsibilities.

03 / ON YOUR PHONE

A file versus a live call

A lab file sent over TCP can recover missing bytes. A real-time voice app often uses UDP-based media and may conceal late or lost audio rather than wait for every packet. The app chooses the right trade-off.

REMEMBER

Layer 4 asks: which app endpoint, and what delivery guarantees?

05

LAYER 5 / OSI REFERENCE MODEL

Session

DIALOGUE & CONTINUITY

THE IDEA, IN PLAIN LANGUAGE

Session establishment, management, and synchronization. This responsibility keeps a conversation organized: when it starts, how it continues, and where it can resume.

STACK CONTEXT

- 7 Application
- 6 Presentation
- 5 Session •
- 4 Transport
- 3 Network
- 2 Data Link
- 1 Physical

DATA UNIT
Session data

SEND ↓ / RECEIVE ↑

01 / HOW IT WORKS

CONCEPTUAL DATA FLOW

Keep track of the conversation

A large upload illustrates session-like state: an identifier and a checkpoint let the service recognize an ongoing task.



FOLLOW THE DATA

The app starts a dialogue, exchanges data with a shared context, and records progress. After an interruption it may restore that context, then close the session.

02 / CORE RESPONSIBILITIES

01 **Establish and end a dialogue**
Create conversation context and release it when the task finishes.

02 **Manage ongoing exchanges**
Track dialogue state and coordinate who sends or responds.

03 **Synchronize and recover**
Use checkpoints to resume supported tasks after interruption.

EXAMPLE PROTOCOLS / STANDARDS

OSI X.225	Formal OSI session protocol.
RPC	Call context; session-like behavior.
SIP	Sets up and ends media sessions.

SIP, RPC, and resumable uploads illustrate session responsibilities at the Internet application layer; they are not literal OSI Layer 5 implementations.

03 / ON YOUR PHONE

Resuming a mobile upload

A student uploads a lab video, then briefly loses coverage. A resumable-upload service keeps the upload ID and last accepted offset. The app reconnects and continues from that checkpoint instead of starting again.

REMEMBER

Layer 5 asks: which conversation is this, and where did we stop?

OSI responsibilities applied to mobile networking — not a literal seven-layer LTE/5G implementation.

06

LAYER 6 / OSI REFERENCE MODEL

Presentation

REPRESENTATION & DATA TRANSFORMATION

THE IDEA, IN PLAIN LANGUAGE

Data formatting, encryption, and compression. This responsibility makes information representable, compact, and protected so the receiving app can interpret it correctly.

STACK CONTEXT

- 7 Application
- 6 Presentation •
- 5 Session
- 4 Transport
- 3 Network
- 2 Data Link
- 1 Physical

DATA UNIT

Transformed data

SEND ↓ / RECEIVE ↑

01 / HOW IT WORKS

CONCEPTUAL DATA FLOW

Same meaning, a different representation

A conceptual sending pipeline transforms content into transferable bytes. The receiver reverses the transformations.



FOLLOW THE DATA

Send: encode content, optionally compress it, then protect the bytes. Receive: decrypt, decompress, and decode before the app uses the content.

02 / CORE RESPONSIBILITIES

01 **Agree on representation**
Use known encodings and formats so both ends interpret bytes alike.

02 **Reduce data size**
Compress content to reduce transfer size and bandwidth usage.

03 **Protect exchanged data**
Encryption hides content; integrity checks reveal modification.

EXAMPLE PROTOCOLS / STANDARDS

UTF-8 / JSON	Text encoding and structured data.
JPEG / gzip	Image and content compression.
TLS	Encryption and integrity for HTTPS.

TLS illustrates presentation-like security, but is not a literal OSI Layer 6 implementation. Internet protocols do not map neatly to all seven layers.

03 / ON YOUR PHONE

Reading a secure API reply

A phone receives a compressed JSON response over HTTPS. TLS verifies and decrypts the records; the HTTP stack decompresses the body; the app decodes UTF-8 JSON into values it can display.

REMEMBER

Layer 6 asks: how is the information represented and protected?

OSI responsibilities applied to mobile networking — not a literal seven-layer LTE/5G implementation.

LAYER 7 / OSI REFERENCE MODEL

07 Application

NETWORK SERVICES FOR APPLICATIONS

THE IDEA, IN PLAIN LANGUAGE

Web, email, video, and mobile apps. This layer defines the messages that apps exchange to request services and present useful information to people.

STACK CONTEXT

- 7 Application •
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network
- 2 Data Link
- 1 Physical

DATA UNIT

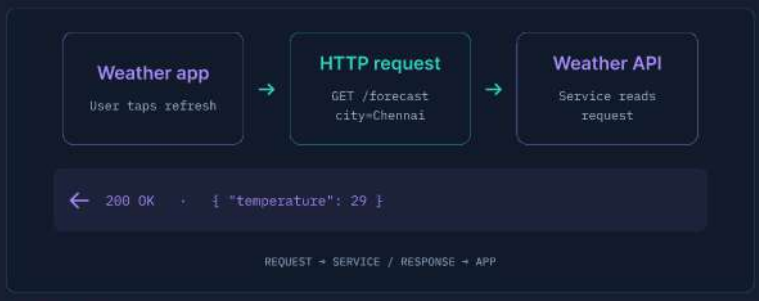
Application data

SEND ↓ / RECEIVE ↑

01 / HOW IT WORKS — CONCEPTUAL DATA FLOW

A request becomes a useful answer

Application protocols give the phone and the remote service a shared language. The lower layers carry those messages.



FOLLOW THE DATA

Send: the app creates a request and hands its bytes to lower layers. Receive: the service interprets the request and returns application data.

02 / CORE RESPONSIBILITIES

01 **Define service messages**
Specify requests, responses, commands, and message meaning.

02 **Access remote resources**
Fetch web content, exchange mail, or locate a network service.

03 **Support user-facing tasks**
Connect app features to services, not directly to radio signals.

EXAMPLE PROTOCOLS / STANDARDS

HTTP / HTTPS	Web pages and APIs; HTTPS adds TLS.
SMTP / IMAP	Sending and retrieving email.
DNS	Resolve service names to addresses.

Apps are not themselves OSI protocols. Video and mobile services can use several protocols across multiple responsibilities.

03 / ON YOUR PHONE

Refreshing the forecast

A student opens a weather app on 5G. The app sends an HTTPS request to a forecast API and reads the returned JSON to show the temperature. Layer 7 supplies the request and response semantics.

REMEMBER

Layer 7 asks: what service or information does the app need?

OSI responsibilities applied to mobile networking — not a literal seven-layer LTE/5G implementation.